CS31001/IT03003 (Software Engineering) 2025

Solution and Answering scheme for the Software Engineering question paper.

Section A: Short Answer Theory

Evaluation Scheme for Section A

Each sub-question is worth 1 mark. Answers should be concise (2-3 sentences each).

a) **Software crisis and causes:** Correct definition gets 0.5 marks; stating causes (e.g., complexity, unreliable performance, schedule delays) adds 0.5 marks.

b) **Prototype Model for Requirement Uncertainty:** Explaining how iterative refinement reduces uncertainty earns full marks.

c) **Documentation in Requirement Engineering:** Listing two reasons such as traceability and communication clarity (0.5 each).

d) **Role of Feasibility Study in Project Initiation:** Justifying decision-making impact (0.5) and mentioning risk assessment (0.5).

e) e. **Sections of IEEE SRS:** Naming sections like Introduction, Functional Requirements, Non-functional Requirements, External Interface Requirements (0.25 each, total 1).

f) **Objectives of SQA Plan:** Listing objectives such as ensuring quality standards, defect minimization, process improvement (0.33 each).

g) **SEI-CMM Maturity Levels:** Listing levels like Initial, Repeatable, Defined, Managed, Optimizing (0.2 each).

h) **Black Box Testing Techniques:** Naming two techniques (e.g., Equivalence Partitioning, Boundary Value Analysis) and giving examples (0.25 each).

i) **Regression Testing:** Correct definition and brief purpose (1).

j) **Purpose of Configuration Management:** Correct statement on version control and tracking changes (1).

Section B: Long Answer and Numerical

Q2. Models and Scrum Plan

a) Spiral vs Iterative for Real-Time Traffic Control

Marks:

- Criteria identification (1),
- Analysis/comparison (2),
- Recommendation (2)

Spiral offers better risk management essential for real-time control; iterative is cost-efficient but may lack detailed risk analysis. Recommend Spiral due to higher risk adaptability.

b) Scrum Plan for 3 Weeks

Marks:

- Team roles (1),
- Backlog selection (1),
- Sprint goals (1),
- Sprint activities sequence (2)

==========

- Roles: Product Owner, Scrum Master, Dev Team.
- Initial backlog: User login, Course creation.
- Goals: MVP dashboard.
- Sprint sequence: Planning, Daily Stand-ups, Development, Review, Retrospective.

Q3. DFD and Decision Table/Tree

a) DFD for Health Monitoring System

Marks:

- Entities (1),
- Processes (2),
- Data Stores (1),
- Data Flows (1)

Create DFD showing Health Worker, Patient, Admin; processes like Registration, Recording; stores for patient/health reports; flows among these.

b) Attendance Decision Table/Tree

Marks:

- Scenarios identification (2),
- System actions in table/tree (2),
- Business rules analysis (1)

List Present, Absent, leave with actions (Mark present, notify absent, approve leave). Discuss how rule-based representation aids in system logic.

Q4. Quality Models and Testing

a) ISO 9000 vs SEI-CMM in Project Management

Marks:

- ISO 9000 analysis (2),
- SEI-CMM analysis (2),
- Integration/barriers (1)

Detail how both foster quality culture, ISO for process, SEI-CMM for maturity. Propose combined adoption challenges.

b) Cyclomatic Complexity Calculation

Marks:

- White-box methodologies (2),
- Cyclomatic metric explanation (1),
- Calculation (2)

Cyclomatic complexity ₹VG = E - N + 2P , $foroneconnectedcomponent$(P = 1 ₹):
₹ VG = 35 - 25 + 2 = 12 ₹.

Q5. Cohesion/Coupling and Design Paradigms

a) Cohesion/Coupling Types

Marks:

- Types (2),
- Importance for design quality (2),
- Object-oriented specifics (1)

Types: functional, sequential, logical cohesion; content, data, control coupling. Discuss high cohesion/loose coupling for maintainable code.

b) Function vs Object-oriented Paradigm

Marks:

- Structural differences (2),
- Encapsulation benefits (3)

Function-oriented: modules, procedures. OO: classes, objects, encapsulation promotes independent development and maintenance.

Q6. Numerical: COCOMO and Function Point Analysis

a) COCOMO Effort and Time Calculation

Project size: 80 KLOC
COCOMO coefficients (embedded): effort=a3.6, b1.20; time=c2.5, d0.32

Effort (PM): ₹ $E = a \times (KLOC)^b = 3.6 \times (80)^{1.20}$ ₹

₹ $(80)^{1.20} \cong 233.64$ ₹

₹ $E = 3.6 \times 233.64 \cong 841.10$ ₹ Person-months

Time (months): ₹ $T = c \times (E)^d = 2.5 \times (841.10)^{0.32}$ ₹

₹ $(841.10)^{0.32} \cong 7.97$ ₹

₹ $T = 2.5 \times 7.97 \cong 19.93$ ₹ Months.

b) Function Point Calculation

FP Elements weighted standard:

- EI: 3 low, 4 average, 3 high

- EO: 5 low, 5 average, 2 high
- EQ: 3 low, 3 average, 1 high
- ILF: 2 low, 4 average, 2 high
- EIF: 2 low, 3 average, 1 high

Apply standard weights (e.g., EI: low=3, avg=4, high=6; EO: low=4, avg=5, high=7; EQ: low=3, avg=4, high=6; ILF: low=7, avg=10, high=15; EIF: low=5, avg=7, high=10):

- EI: $3 \times 3 + 4 \times 4 + 3 \times 6 = 9 + 16 + 18 = 43$
- EO: $5 \times 4 + 5 \times 5 + 2 \times 7 = 20 + 25 + 14 = 59$
- EQ: $3 \times 3 + 3 \times 4 + 1 \times 6 = 9 + 12 + 6 = 27$
- ILF: $2 \times 7 + 4 \times 10 + 2 \times 15 = 14 + 40 + 30 = 84$
- EIF: $2 \times 5 + 3 \times 7 + 1 \times 10 = 10 + 21 + 10 = 41$
- UFP = 43 + 59 + 27 + 84 + 41 = 254

Adjustment Factor: CAF = 0.65 + (0.01 × 50) = 1.15 (high complexity)
Adjusted FP: AFP = UFP × CAF = 254 × 1.15 = 292.1